ReCycleGuard

An audited agentic assistant for e-waste EPR reconciliation

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The problem

- Under the E-Waste (Management) Rules, 2022, a producer must meet a yearly EPR obligation (60%, 70%, then 80% of a historical sales cohort) with certificates from registered recyclers.
- Obligations, sales, purchases, payments, portal transfers and balances are *different facts in different records*; a payment does not prove a transfer.
- Reconciliation is done by hand and fails in known ways: double counting, wrong entity, mixed units, rules from the wrong period, expired certificates, unreadable pages.

Decision supported

For this producer and period: which obligation balances are supported by verified evidence, which discrepancies need follow-up, and what must a human reviewer confirm?

Why an agent, and what stays deterministic

Needs an agent

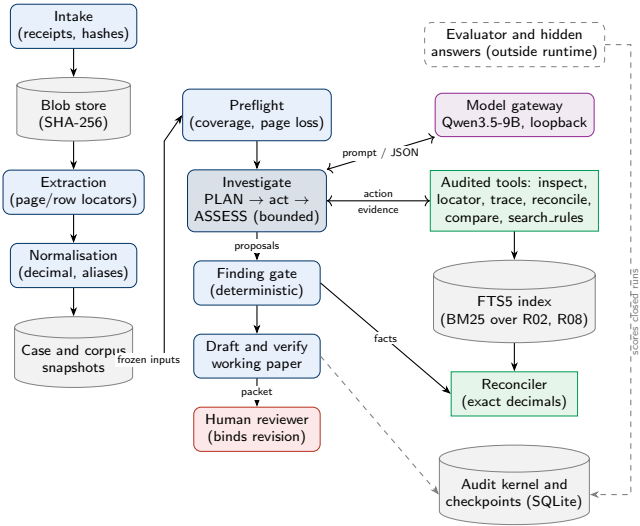
- The next check depends on what the last one showed
- Long rules and dossiers: retrieval, not one prompt
- Weak evidence must lead to another check or an explicit abstention

Must not be the model

- Balances: exact decimal arithmetic in code
- Accepting a finding: a deterministic gate
- Approving the paper: a human reviewer

So the project tests the agentic hypothesis instead of assuming it: model-free controls run through the same pipeline in every session.

Architecture



The bounded investigation loop

- LangGraph state graph: **preflight** → **investigate** → **draft** → **verify**, checkpointed in SQLite with compare-and-swap versions.
- Each turn: PLAN proposes *one* audited tool call (or stops with a reason); the tool runs on the frozen snapshot; ASSESS returns a verdict and resolve / continue / abstain. Resolve requires *supports*.
- Six tools: inspect evidence, locator lookup, certificate trace, reconciliation, return comparison, rule search (BM25 over the Rules and the CPCB FAQ, graded).
- Bounds: 12 model calls, 600 s, 32,768 tokens per call; repeated actions refused; invalid output retried once, then deterministic evidence only.
- Model: Qwen3.5-9B Q8_0 on llama.cpp b12033, local, loopback only.

Agentic concepts, with evidence

Concept		Evidence
Tool use	✓	six audited tools; every call recorded before dispatch
Agentic RAG	✓	graded rule search; MG2 Retriever grades results and must re-query
Planning and reasoning	✓	PLAN/ASSESS loop; fixed (V1) vs adaptive (V2)
Observability	✓	per-trial ledger, result records, audit-chain check
Evaluation	✓	deterministic scorer, hidden answers, controls, bootstrap CIs
Multi-agent (bonus)	✓	Retriever + Classifier, pre-registered, measured
Human in the loop, guardrails	✓	finding gate; review bound to the report revision
Durable workflow	✓	checkpoints and resume; queue resumes from ledgers
Experimentation	✓	MG1 chain, MG2 experiment, one held-out run
Memory	partial	within a run only, by design (no leakage between cases)

How it was evaluated

- **48 synthetic dossiers** (`pilot-0.1`), sixteen fault families F01–F16, one per family per split plus combinations; grouped splits, similarity checks.
- **Splits:** tuning-16 (decisions), checking-8 (looked at once), held-out 24 (sealed, run once). **Ablated arena**: tuning-16 with every deterministic family check off.
- **Primary metric:** supported case success: exact buckets, exactly the expected finding families, required scopes withheld, no forbidden action.
- **Controls in every session:** V0 (no model) and SC (scripted controller). If a model arm cannot beat SC, the model adds nothing measurable.
- Hidden answers never enter the runtime; one Docker container per trial on the PC.

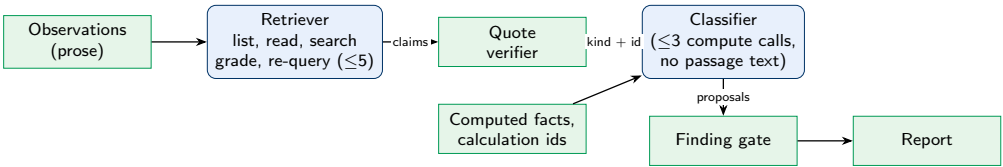
MG1: the single agent

- Prompt tuning plateaued: v7 succeeded on 69 of 128 tuning trials; the model contributed 3 findings.
- Tracing the failures showed deterministic causes, so the *engine* was fixed: family checks and a contradiction gate took v7 to 30, then 32 of 32.
- **But the controls matched it**, and the model's share of findings was 0.
- With every family check off: V1, V2 and SC all 0 of 21 findings; a thinking prompt found 1 with false claims and regressed to 26 of 32 with the rules on.

Reading

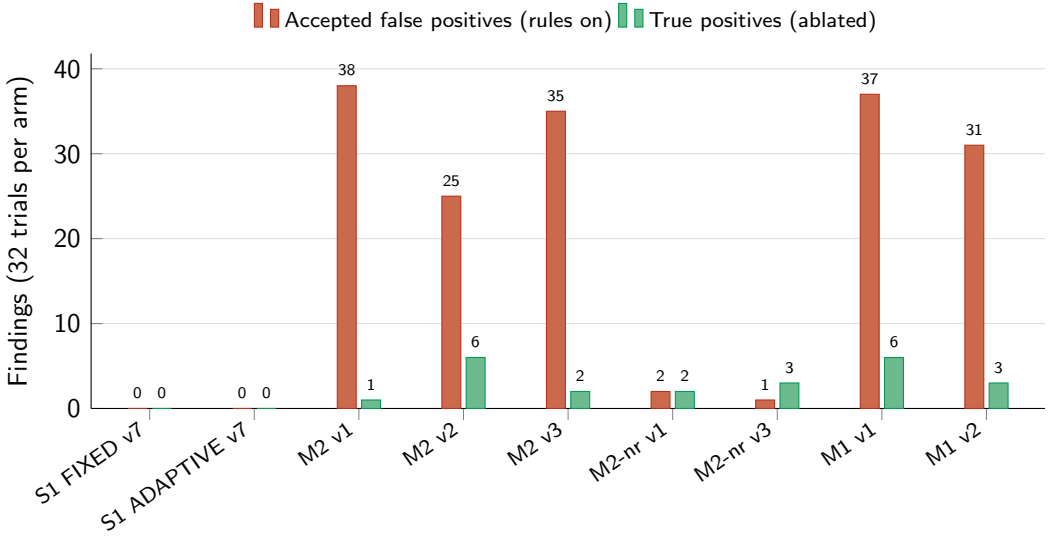
The gains came from the deterministic engine. The model never read document text.

MG2: two agents, pre-registered



- **Hypothesis:** a Retriever that reads and grades text, plus a Classifier over verified claims and computed facts, recovers judgement faults in the ablated arena without adding false findings when the rules are on.
- Arms M2 (both), M1 (Retriever + lookup), M2-nr (Classifier alone) against S1 (v7) and SC; prompts v1–v3 declared in advance; selection rule fixed before the first run.
- Ran on free Kaggle T4 GPUs as a separate series while the PC was down: 18 sessions, 1,008 trials, scored on the laptop.

MG2 result: cost and benefit per arm



No arm both recovers faults with the rules off and stays clean with them on.

MG2 result: the rules-on gate

MG2 rules-on arena (the no-regression gate)

Arm	Prompt	Cases	95% CI	TP / FP	Prop. / acc.	Calls	s
S1 v7 FIXED	v7	32 of 32	1.00–1.00	42 / 0	1 / 0	6.2	107
S1 v7 ADAPTIVE	v7	31 of 32	0.91–1.00	41 / 0	1 / 0	2.3	90
M2 Retriever + Classifier	v1	10 of 32	0.16–0.47	41 / 38	48 / 38	5.7	87
M2 Retriever + Classifier	v2	15 of 32	0.31–0.62	42 / 25	38 / 25	5.3	134
M2 Retriever + Classifier	v3	12 of 32	0.19–0.59	40 / 35	46 / 35	5.7	136
M2-nr Classifier alone	v1	30 of 32	0.81–1.00	42 / 2	26 / 2	1.1	51
M2-nr Classifier alone	v3	31 of 32	0.91–1.00	42 / 1	18 / 1	1.1	49
M1 Retriever + lookup	v1	9 of 32	0.12–0.47	41 / 37	45 / 37	4.8	100
M1 Retriever + lookup	v2	13 of 32	0.25–0.59	41 / 31	44 / 31	4.4	90
SC (in each of 9 sessions)	–	16 of 16, FP 0					
V0 (in each of 9 sessions)	–	16 of 16, FP 0					

T4 series, cooperative isolation, 3 slots, 2 repeats on tuning-16. Prop. / acc.: model finding proposals made and accepted by the gate. Calls and s: mean model calls and wall seconds per trial. Source: `docs/checkpoints/mg2/runs/t4/*/compare.csv`.

Held-out run (24 sealed dossiers, once)

Declared before it ran (log entry 15): V1 vs V2 with v7 (primary comparison), V0 and SC, plus the Classifier alone v3 (descriptive). Freezes pin commit, protocol, prompts and dossier bytes.

Held-out run (24 sealed cases, rules on, 2 repeats, T4)

Arm	Prompt	Cases	95% CI	TP / FP / FN	Prop. / acc.	Calls	s
S1 V1 FIXED	v7	48 of 48	1.00–1.00	64 / 0 / 0	3 / 0	6.1	83
S1 V2 ADAPTIVE	v7	48 of 48	1.00–1.00	64 / 0 / 0	3 / 0	2.4	73
SC (S1 session)	–	24 of 24	1.00–1.00	32 / 0 / 0	0 / 0	2.0	27
V0 (S1 session)	–	24 of 24	1.00–1.00	32 / 0 / 0	0 / 0	1.0	26
M2-nr Classifier alone	v3	47 of 48	0.94–1.00	64 / 1 / 0	31 / 1	1.1	42
SC (M2-nr session)	–	24 of 24	1.00–1.00	32 / 0 / 0	0 / 0	2.0	29
V0 (M2-nr session)	–	24 of 24	1.00–1.00	32 / 0 / 0	0 / 0	1.0	28

Run once after the freeze (log entry 15). M2-nr is descriptive: it failed the development gate. Source: `docs/checkpoints/mg2/runs/t4/mg2-t4-heldout/compare.csv`.

V1 and V2 tie at the ceiling with both controls, so the engine generalises to these cases; V2 needs 2.4 model calls against 6.1. The model's accepted findings: none.

Decision and what we learned

- **No multi-agent arm passed the gate.** Every Retriever arm accepted 25–38 false findings in 32 trials, mostly F05 (wrong entity alias) and F14 (out-of-scope regime), whatever the prompt.
- The Classifier alone was nearly clean only because the gate rejected almost everything it proposed.
- Prompt iteration stopped after three versions; the remaining budget went to one honest held-out run, where every loop and control succeeded on all 24 unseen cases.
- At this model size, the value is in the deterministic tools, the gate and human review. The agent loop's demonstrated property is that it is *bounded, traceable and safe to fail*.

Limitations and responsible use

Limitations

- Synthetic dossiers from one generator, built by the same team that built the system
- Small sets; wide intervals
- One 9B model only
- T4 runs used process-level isolation
- Why the gate accepts F05/F14 claims is not yet analysed

Responsible use

- Advisory working paper only: no filing, trading or compliance verdict
- Every finding cites evidence and calculations
- A person reviews; the decision binds the exact revision
- Local inference; no real producer data

Conclusion and future work

- Built: an audited evidence pipeline, a bounded LangGraph agent with a deterministic gate and human review, a 48-dossier benchmark with an isolated evaluator, and an unattended trial harness for a GPU PC or a free cloud T4.
- Measured: honest negative result for the model, positive for the design around it.
- Next: analyse and close the gate's F05/F14 gap; rerun MG2 with a larger model; an independently authored dossier set; more sources; held-out rerun with container isolation.

Repository: `https://github.com/R-V-Abhishek/ReCycleGuard` Report: `docs/report/report.pdf`